



NIS16 — CTTHEES2 Milestone Progress Report

Thesis: *Cross-Layer Dataset Design and Exploratory Analysis of ESP32-Based ESP-WIFI-MESH Network* **Institution:** De La Salle University · College of Computer Studies **Report date:** 2026-07-23 **Prepared for:** Proponents (Calpoporo, Carlos, Ong, Reinante) · Adviser: Cu, Gregory G.



TL;DR — Where We Stand

Status		Milestones
● Done	Fully meets criteria	M1 Firmware · M6 Preprocessing
● Partial	Substance done, criteria gaps	M2 Attacks · M3 Topologies · M5 Extraction · M7 Features · M8 EDA
● Not yet	Biggest open gap	M4 Experiment repeats (≥ 24 runs)

🔧 **Three things to close before final demo:** 1. ● **Repeats (M4)** — run ≥ 3 per cell to reach the ≥ 24 -run matrix 2. ● **Topology coverage (M3)** — fill the 3 missing attack $\times$ topology cells 3. ● **Uniformly-NaN features (M7)** — firmware fix or documented exception with adviser

? Q&A — The Null-Value Question (answered)

Your question: "Is it normal to have null values in some columns/rows in the EDA tables? Wormhole doesn't have a supposed value in PDR, right?"

✅ **Short answer: Yes — NaN cells are normal, expected, and documented. Not a bug.**

⚠️ One correction to the framing

It's **not** that "wormhole doesn't have PDR." Wormhole runs **do** have PDR — it's populated for the **victim** rows (the probe senders). Verified directly against real data:

```
analysis/wormhole/linear_topology/feature_table.csv
• 723 total rows
• PDR is NaN on 268 rows → exactly the NON-victim rows
  (root + the two wormhole tunnel endpoints)
• roles: victim=284, wormhole_b=171, wormhole_a=136, root=132
```

🧩 The three legitimate reasons a cell is NaN

Type	Features affected	Why it's correct
👤 Role-specific	<code>PDR</code> (victims only) · <code>Tunnel*</code> (wormhole endpoints only) · <code>ForwardingRatio</code> / <code>IngressEgressDelta</code> / <code>ConsistencyScore</code> (blackhole attacker only)	A forwarding ratio is undefined for a node that doesn't relay; PDR is undefined for the root itself. NaN ≠ zero.
🦋 Attack-specific	<code>TunnelIntensity</code> · <code>TunnelBytes</code>	Only exist in wormhole runs — NaN in baseline & blackhole entirely. Thesis Table 4.12 : "null or zero for all other nodes and phases."
🔧 Firmware-blocked	<code>LatencyHopRatio</code> · <code>TunnelLatency</code> (+ currently the 3 forwarding features)	No RTT round-trip leg exists (probes are one-way), so latency can't be timed. Flagged as NaN rather than faked.

🚨 One caveat to remember for the demo

`PDR = 0.0` vs `PDR = NaN` are **different meanings**: - 🟡 `0.0` during an attack window = the **real blackhole signature** (root heard the node, but zero probes arrived). - 🟠 `NaN` = the root **never heard that node at all** (coverage gap, wrong topology, CSV not pulled).

👉 Don't let these two get conflated in a plot — your code (`features.py`) already separates them correctly, keep it that way.

✅ Milestone-by-Milestone Checklist (CTTHES2 · 100%)

Legend: 🟢 Done · 🟡 Partial · 🟠 Not yet

🟢 M1 — Firmware, All Node Roles · 15% · DONE

- ✅ `mesh_setup.c` , `phase_listener.c` , `csv_logger.c` — all present in `components/mesh_common`
- ✅ `root_main.c` + `victim_main.c` implemented
- ✅ Baseline runs complete; CSVs retrievable via USB serial
- 🟢 **Meets all criteria.**

🟡 M2 — Application-Layer Attack Modules · 15% · MOSTLY DONE

- ✅ `blackhole_victim.c` — silent-drop attack, drop signature captured
- ✅ `wormhole_victim.c` — confirmed working on **COM26/COM27** (~180-probe duplicate signature at root)
- ⚠️ Criterion "behavior consistent across all four topologies" **not yet demonstrated** → see M3

● M3 — Multi-Topology Deployment · 15% · PARTIAL

Topology	Baseline	Blackhole	Wormhole
🌲 Tree	✓	✓	✓
— Linear	✓	✓	✓
★ Star	✓	✓	●
Partial mesh	✓	●	●

- **Missing cells:** star-wormhole · partial-mesh-blackhole · partial-mesh-wormhole
- ⚠️ **Data-quality flag:** linear mesh is unstable (weak 5-hop chain, re-parenting, ROOT LOST). Cleaner data → re-run with better board spacing and/or `-Topology tree`.

● M4 — Phase-Controlled Experiment Execution · 15% · BIGGEST GAP

- 📄 **Requirement:** ≥ 24 runs = 4 topologies × 2 attacks × ≥ 3 repeats
- **Current state:** every run is `r1` only (single repeats) — exploratory, not 3× replicated
- 👉 **This is the milestone furthest from its criterion.** Priority #1.

● M5 — Raw Extraction & Integrity Validation · 10% · PARTIAL

- ✓ Extraction works — `run.ps1 -Export` pulls telemetry + arrivals CSVs
- ⚠️ Formal validation report "confirms ≥ 24 clean runs" **cannot complete until M4 produces 24 runs**
- 🔍 **TODO:** confirm `m5_extraction/` has the coverage ($\geq 95\%$) / label / corruption validator wired up

● M6 — Preprocessing Pipeline · 10% · DONE

- ✓ `preprocess.py` : per-node re-basing · cumulative→delta · 5s windowing · low-sample discard · modal labels
- ✓ Deterministic · folder-in → unified windowed DataFrame out
- **Meets all criteria.**

● M7 — Feature Engineering (16 features) · 10% · STRUCTURALLY DONE, ONE REAL GAP

- ✓ All 16 columns present · 11 compute real values
- ⚠️ Criterion "no feature is uniformly NaN" **not fully met:**
- `LatencyHopRatio` and `TunnelLatency` are **NaN in every run** (no RTT leg in firmware)
- `ForwardingRatio` / `IngressEgressDelta` / `ConsistencyScore` only populate on the blackhole attacker
- 🔧 **Decision needed:** (a) add response leg / recv-forward counters to firmware, **or** (b) get adviser sign-off that these are intentionally-flagged NaN per Table 4.12

● M8 — Exploratory Data Analysis · 10% · MOSTLY DONE, QUALITY-LIMITED

- ✓ All 5 analyses in `eda.py` : descriptive stats · distributions · Pearson/Spearman correlation · PCA/t-SNE
- ✓ Outputs generated for 9 combos

- ⚠️ Criterion needs ≥ 1 feature clearly separating baseline vs each attack **with quantitative backup** — plots exist, but separation quality is limited by the mesh-stability + single-repeat issues above

Data Coverage Snapshot (as captured)

Feature tables present (9):

```
baseline : linear · partial_mesh · star · tree
blackhole: linear · star · tree
wormhole : linear · tree
```

Raw telemetry exports ([tools/exports/](#)):

```
baseline/linear_topology      8 files
baseline/partial_mesh_topology 4 files
baseline/tree_topology        4 files
blackhole/linear_topology     7 files
blackhole/star_topology       4 files
blackhole/tree_topology       8 files
wormhole/linear_topology      5 files
wormhole/tree_topology        4 files
```

Repeat check: all runs are **r1** — confirms the M4 replication gap. ●

Recommended Next Actions (priority order)

1. 🏆 **Hit the run matrix (M4)** — schedule ≥ 3 repeats per attack×topology cell; capture across time-of-day for interference variation.
 2. 🧩 **Fill missing topology cells (M3)** — star-wormhole, partial-mesh-blackhole, partial-mesh-wormhole.
 3. 🛠️ **Resolve uniformly-NaN features (M7)** — firmware change vs. documented exception; get it in writing from the adviser.
 4. 📏 **Stabilize the linear mesh** before treating its data as final — better spacing or fall back to tree.
 5. 📄 **Wire up / verify the M5 validation report** so it's ready to certify the 24 clean runs once they exist.
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Reference — What Each Feature Needs

Feature	Source	Status
RetryRate , RSSI_mean/var/stability , RSSI_Hop_Diff , ParentSwitchRate , LayerChangeCount , HopStabilityDuration	Logged directly by firmware	● real
PDR	Root arrivals ∞ victim probes on (src_mac, seq_num)	● real (victims)
TunnelIntensity , TunnelBytes	Wormhole endpoint counters	● attacker-only
ForwardingRatio , IngressEgressDelta , ConsistencyScore	Needs recv/forward counter split	🔑 firmware-blocked
LatencyHopRatio , TunnelLatency	Needs RTT round-trip leg (none exists)	🔑 firmware-blocked

🧩 Overall: strong analysis pipeline, firmware & attacks working — the remaining work is **volume (repeats) + coverage (topologies) + one firmware/feature decision.**
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